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Portfolio: weirdowrench.github.io | linkedin.com/in/giridhera-ramanan
github.com/WeirdoWrench

Profile Summary

Final-year AI & Data Science student with hands-on experience in full-stack AI applications, computer vision, and IoT systems. Seeking roles in Generative AI engineering and full-stack development, with a focus on LLM-powered applications.

Education

Animalar Engineering College Chennai City Campus, Nov 2022 – May 2026
B.Tech Artificial Intelligence & Data Science **CGPA: 8.35/10**

Technical Training

Java Development Trainee, June 2025 – Aug 2025
Robowaves - IN
• Solved 100+ logical programming problems and practical exercises focusing on Data Structures, Arrays, String Manipulation, and the Collections Framework.

Experience

IoT Intern, Nov 2024 – Jan 2025
KUBE RESEARCH LABS PRIVATE LIMITED - Chennai, IN
• Developed and integrated BLE-based RSSI tracking and ultrasonic sensor logic with Arduino Uno for an autonomous smart cart, achieving reliable real-time object avoidance within a 30cm detection range.
• Designed and executed a 50+ test validation protocol for autonomous cart prototypes, reducing hardware fault rate and driving measurable improvements in navigation stability and sensor response time.

Projects

Aura – AI-Powered Mental Health Journal aura-journal.netlify.app
Next.js, FastAPI, PostgreSQL (Neon), Llama 3.1 / Gemma 2, Groq, Tailwind CSS
• Architected a full-stack mental wellness platform using Next.js and FastAPI, deployed on Vercel and Render with optimized Neon Postgres connection pooling to reduce database geographic latency by over 90%.
• Engineered a high-availability AI routing engine utilizing Groq, TogetherAI, and OpenRouter to seamlessly fall back across Llama 3.1 and Gemma 2 models, ensuring 100% uptime for real-time empathetic chat and sentiment analysis.
• Designed a 3-tier enterprise security pipeline featuring database encryption, an automated regex PII scrubber to anonymize data in transit, and SlowAPI rate-limiting with active prompt-injection defense mechanisms.
• Developed a responsive, glassmorphism-styled frontend using Tailwind CSS, integrating the Web Speech API for voice dictation and Recharts for dynamic mood trend visualization and AI-generated personality profiling.
• Implemented interactive community features including anonymous posting, hashtag-based feed filtering, and a gamified achievement badge system to drive user retention.

Accident Detection with CNN & MobileNetV2 github.com/WeirdoWrench/RAD
TensorFlow, OpenCV, SMOTE, ImageDataGenerator
• Developed a real-time accident detection pipeline using a MobileNetV2-based CNN with TensorFlow, applying SMOTE to handle class imbalance in the training dataset and achieving 82.7% validation accuracy and 84.3% precision — outperforming a non-augmented baseline by ~11%.
• Calibrated model confidence thresholds to optimize real-world functional deployment, minimizing false predictions during live inference.
• Integrated OpenCV to process continuous video streams, executing automated frame-by-frame predictions and triggering immediate visual alerts for emergency monitoring.

Portfolio AI Chatbot weirdowrench.github.io
FastAPI, LangChain, Groq, Llama 3.1, Render
• Built and deployed a production AI assistant that answers recruiter queries about skills, projects, and experience in natural language using the resume as live context.
• Engineered a FastAPI backend that parses a PDF resume at startup via PyPDF and constructs a LangChain LCEL chain with Groq's Llama 3.1-8b-instant for sub-second inference.
• Eliminated vector store dependency by leveraging Llama 3.1's 128k context window, reducing memory footprint to under 512MB and enabling free-tier cloud deployment on Render.

- Embedded a custom vanilla JS chat widget into the portfolio with CORS-secured cross-origin API calls, real-time typing indicators, and directional chat bubbles.

Moodflix – Movie Discovery App

moodflix04.netlify.app

React 19, Vite, Tailwind CSS, Appwrite, TMDB API

- Built a responsive movie discovery web app using React 19 and Vite, integrating the TMDB API to browse thousands of movies with real-time search across a dark-themed, mobile-optimised UI.
- Leveraged Appwrite as a BaaS to persist and aggregate user search metrics, powering a dynamic trending section that surfaces the top 5 most-searched movies across all sessions.
- Implemented graceful error handling and empty-state fallbacks for missing posters and failed API requests, ensuring a smooth experience across mobile, tablet, and desktop viewports.
- Engineered a debounced search mechanism with a 500ms delay, reducing TMDB API calls by ~80% during active typing, and implemented lazy-loaded movie cards for improved rendering performance.

Certifications

Java Programming for Beginners - Simplilearn & SkillUp

AWS Academy Cloud Foundations – AWS Academy Graduate

Fundamentals of Machine Learning & Deep Learning – NPTEL Online Certification

Artificial Intelligence for Real-World Applications – TCS iON

Journey to Cloud: Envisioning Your Solution – IBM SkillsBuild

Introduction to Cybersecurity – Cisco

Technologies

Languages: Java, Python, JavaScript, TypeScript, SQL, Bash

Frameworks/Libraries: Node.js, React.js, Next.js, OpenCV, YOLO, TensorFlow (tf.keras), PyTorch, Pandas, Scikit-Learn

Developer Tools: Git, Github, Docker, AWS, Firebase, GNU/Linux